

Sunya Afrasiabi

email | personal website | linkedin

Education

- University of Pennsylvania**, BSE in Mechanical Engineering, BA in Philosophy Sept 2024 - May 2028
- **Relevant Coursework:** Thermal-Fluids Engineering, Statics & Strength of Materials, Thermodynamics, Dynamics, Computer-Aided Design (CAD), Computer Programming (Python), Linear Algebra, Differential Equations, Probability
- University at Buffalo**, Gifted Math Program Sept 2023 - April 2024
- **Relevant Coursework:** College Calculus III, Introduction to Linear Algebra

Experience & Leadership

- University of Pennsylvania – Li Group**, Undergraduate Researcher May 2026 - Present
- Research insect-wing-inspired morphing structures, modeling compliant geometries in **Rhino** and **Grasshopper** to study parametric mechanical behavior.
 - Translate computational designs into multi-material prototypes using **powder-based 3D printing** for experimental validation.
- NASA L'SPACE Proposal Writing & Evaluation Experience**, Radiation Subteam Lead Sept 2025 - Dec 2025
- Developed innovative solution to the limitation of radiation-resistant materials in HDPE composites to enhance material performance under lunar radiation conditions
 - Gained experience in technical proposal composition, project evaluation, and NASA's mission review process.
 - Ranked in the **Top 6 out of 40+ national teams**.
- Penn Astronomical Student Association**, Outreach & Marketing Director Sept 2025 - Present
- Lead outreach initiatives and external partnerships, securing **\$200** in sponsorship funding.
 - Configure and calibrate equatorial telescopes and optical tracking systems for public observation events.
- #MyStory Program**, Founder and Project Manager Jan 2021 - Aug 2024
- Founded and managed a non-profit oral history initiative promoting cultural awareness through exhibitions.
 - Secured over **\$24,000** via targeted donations, fundraisers, and corporate grant programs (including Riley's Way Foundation, Prudential, Ashoka, and the Peter & Elizabeth Tower Foundation).
 - Directed operations for **20+ community partners** and **40+ volunteers**.

Technical Projects

- Multi-Phase Butane Rocket & Numerical Analysis** Spring 2026
- Designed and optimized a variable-mass butane-powered water rocket using aerodynamic modeling, thermodynamic analysis, and experimental validation to navigate constrained flight paths and minimize target error.
 - Developed a custom Python Euler integration solver coupling butane vaporization, pressure-driven propulsion, drag, and ballistic dynamics; implemented trajectory reconstruction and multi-variable optimization to improve predictive accuracy.
- Mars Rover Chassis Prototype** Spring 2026
- Designed and optimized a Mars rover chassis prototype for University Rover Challenge (URC) constraints as part of a 3-person engineering team, developing detailed SolidWorks assemblies and structural architectures.
 - Performed FEA-driven design iterations to optimize load paths, bracket placement, wall thicknesses, and lightweight geometries while balancing structural integrity and mass reduction.
- Vacuum-Powered Agricultural Seed Cleaner** Fall 2025
- Designed and fabricated a vertical pneumatic seed-chaff separator with a zig-zag separation column, custom swirl chamber, laser-cut acrylic panels, and 3D-printed nozzles to optimize airflow behavior and separation efficiency.

Papers

- Linear Algebra Foundations of Neural Networks**, University of Pennsylvania August 2025
- Wrote expository research paper connecting linear algebra, neural networks, and robotics. **(Link)**
- Aviation Solutions for Wildlife Prevention**, Wharton Undergraduate Aerospace Club May 2025
- Co-authored investor-oriented whitepaper on UAVs, AI wildfire prediction, and aerospace innovations, framing \$6.4B+ market opportunities and potential \$5B annual cost savings. **(Link to File)**

Skills

Software: Rhino 3D, Grasshopper, SolidWorks, MATLAB, Python, NumPy, FEA
Fabrication: Laser Cutting, FDM/SLA 3D Printing, Mechanical Assembly
Additional: Computational Geometry, Bio-Inspired Design, Technical Writing, Logic, Project Management